GAD vs Sella on HIP — 3-method benchmark, tuned-Sella canonical

2026-04-28 v2 (revision of 04-28). Transition1x train, 6 noise levels, 2000 steps, IRC = `sella_hip`.

Canonical Sella in this revision = "libdef" tuning config (δ_0=0.1, γ=0.4). It dominates the previously-canonical "default" (δ_0=0.048, γ=0) at 100/150/200 pm by +7–10 pp. Tuning grid that established this is in the appendix (§3.1); GAD-with-larger-dt partial sweep also there (§3.2).

Contents

1 Headline **2**

2 GAD step-size over time **4**

3 Appendix **5**

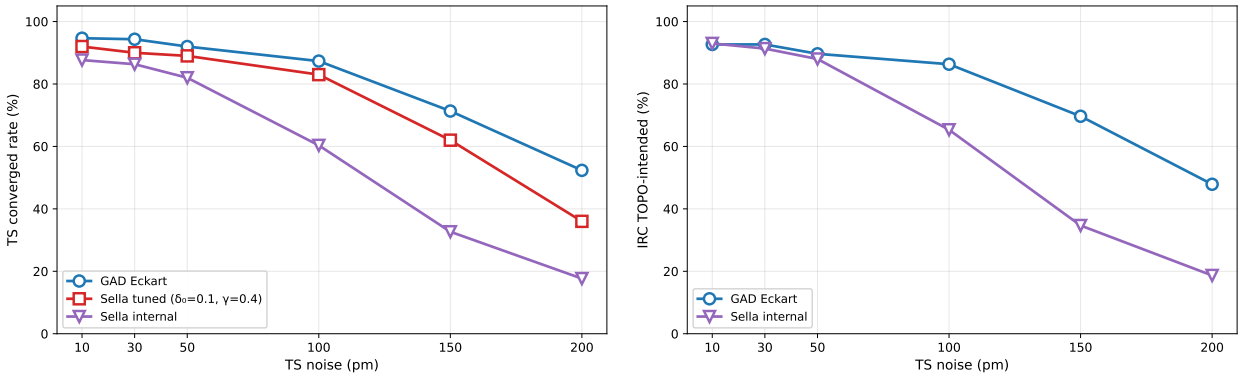
3.1 Sella tuning grid (full results, n=100) 5

3.2 GAD with larger dt (partial) 5

3.3 Data locations 6

3.4 In flight at write time 6

1 Headline



(a) TS-converged rate vs. noise. GAD: n=300. Sella tuned: n=100. Sella internal: n=300.

(b) IRC TOPO-intended rate vs. noise. "Sella tuned" line will appear once IRC sweep on libdef pool lands (job 60003770).

Figure 1: TS-converged rate (left) and IRC TOPO-intended (right) for the three canonical methods.

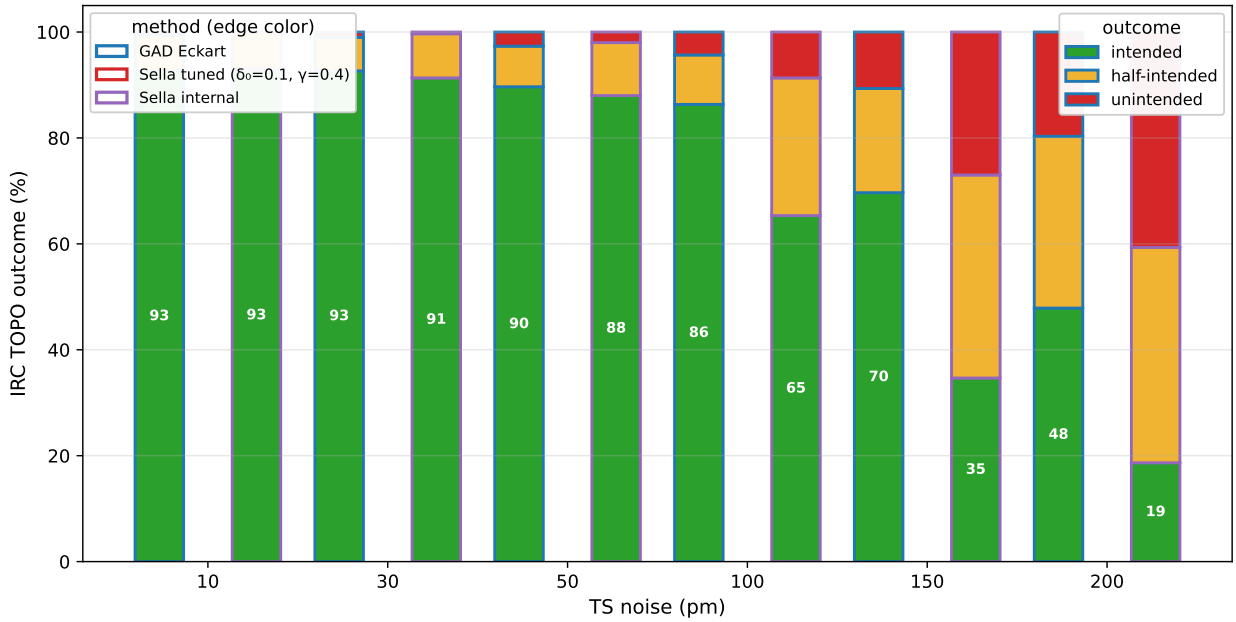


Figure 2: IRC TOPO outcome breakdown per method per noise level. Bar fill = outcome class (intended/half/unintended). Bar edge color = method identity (matches Fig. 1).

TS-converged rate (% by canonical criterion: $n_{\mathbf{neg}}$=1 $\wedge$ $f_{\max}$<0.01)

Method	10	30	50	100	150	200	n
GAD Eckart (dt=0.003)	87.3	87.3	85.3	80.0	63.7	45.3	300
Sella tuned (libdef)	92.0	90.0	89.0	**83.0**	**62.0**	**36.0**	100
Sella internal	87.7	86.3	82.0	60.3	32.7	17.0	300

IRC TOPO-intended rate (%)

Method	10	30	50	100	150	200	n
GAD Eckart (dt=0.003)	93.0	93.0	90.7	86.7	70.0	47.0	300
Sella tuned (libdef)	*IRC sweep pending (job 60003770)*						100
Sella internal	93.0	91.3	88.0	65.3	34.7	18.7	300

2 GAD step-size over time

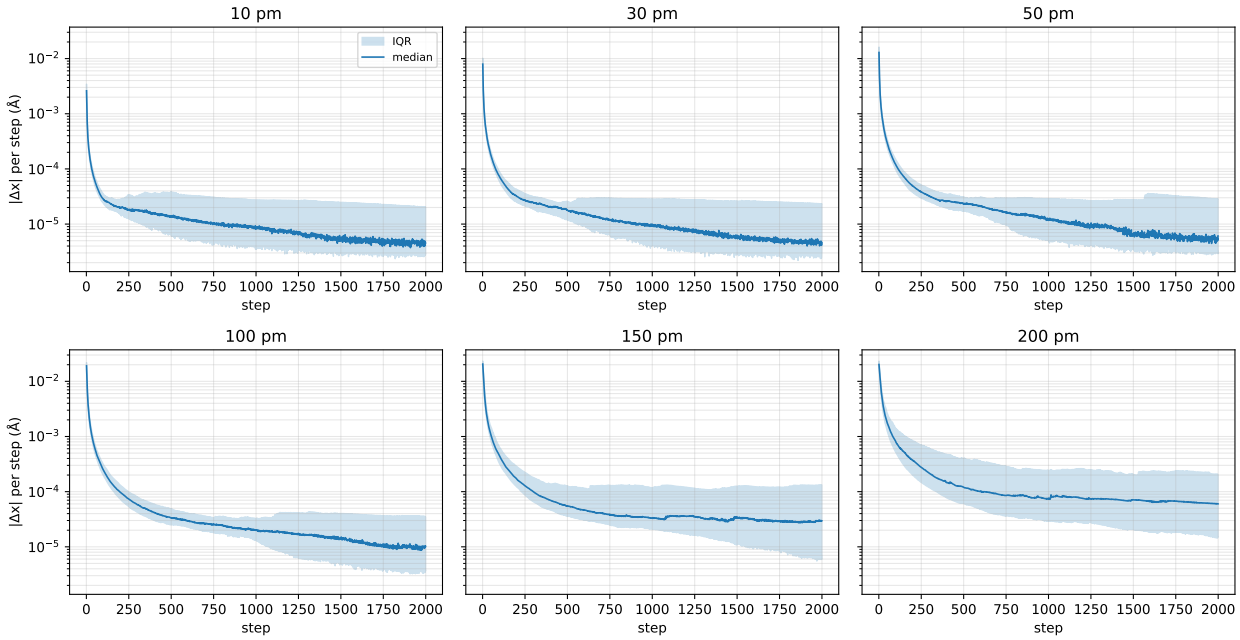


Figure 3: GAD Eckart (dt=0.003) per-step Cartesian displacement (Å). Median (line) and IQR (band), aggregated over 300 trajectories per noise level. Sella step-size companion figure pending Sella traj-log job 60002542.

3 Appendix

3.1 Sella tuning grid (full results, n=100)

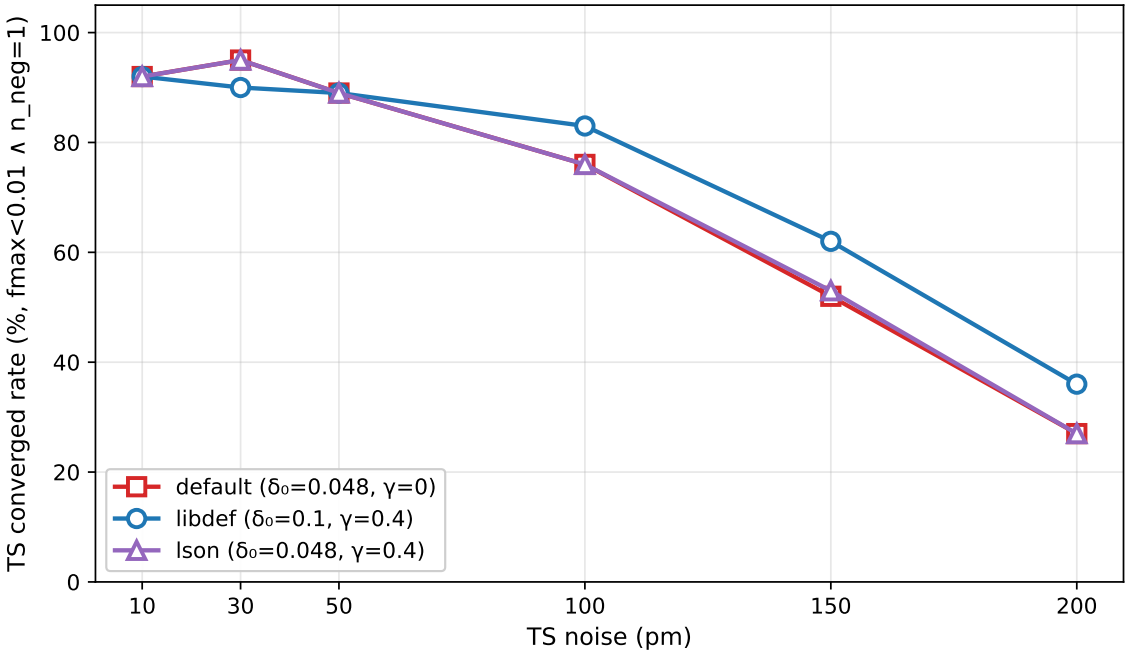


Figure 4: Sella TS-converged rate as a function of two parameters: δ_0 (initial trust radius) and γ (line search; γ=0 disables it). All cart+Eckart, n=100 samples per cell, fmax¡0.01.

Sella tuning, TS-converged rate (%, n=100 each)

Config (δ_0, γ)	10	30	50	100	150	200	mean nfev
default (0.048, 0)	92.0	**95.0**	89.0	76.0	52.0	27.0	589
libdef (0.1, 0.4)	92.0	90.0	89.0	**83.0**	**62.0**	**36.0**	**502**
lson (0.048, 0.4)	92.0	**95.0**	89.0	76.0	53.0	27.0	585

Findings: (1) line search alone (`lson` vs `default`) is inert; (2) larger trust radius (`libdef`) gains +7–10 pp at 100–200 pm and is also *cheaper* (502 vs 585–589 mean function evaluations) because larger trust → fewer steps; (3) the only place where `libdef` loses is 30 pm (90 vs 95) — small enough to potentially be sampling noise at n=100.

3.2 GAD with larger dt (partial)

Sweep in flight (job 59995949): three larger-dt configs ($dt \in \{0.005, 0.010, 0.020\}$) at 6 noise levels, n=300 each, fmax¡0.01. Motivated by §2: GAD per-step displacement at dt=0.003 is $\sim 10^{-3}$ Å, an order of magnitude below Sella's QN trust step.

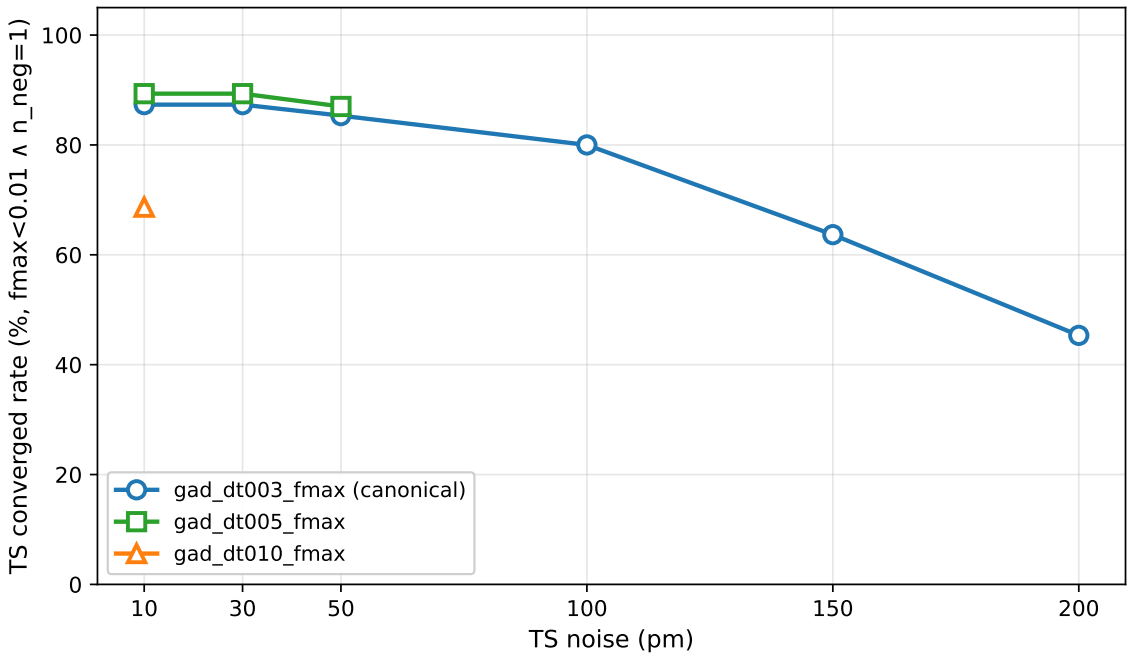


Figure 5: GAD TS-converged rate vs. noise for four step sizes. Cells with no data: sweep still running.

3.3 Data locations

Artifact	Path
GAD Eckart canonical (dt=0.003)	`runs/gad_eckart_fmax/`
GAD bigger-dt sweep (in flight)	`runs/gad_bigger_dt/`
Sella tune grid	`runs/sella_tune/{default,libdef,lson}/`
Sella tune IRC (in flight)	`runs/irc_sella_libdef/`
Sella internal canonical	`runs/sella_2000/`
GAD IRC	`runs/irc_sellahip_allendpoints/`
Sella internal IRC	`runs/irc_sella_int_2000/`
This report	`IRC_COMPREHENSIVE_2026-04-28v2.tex/pdf`

3.4 In flight at write time

- Job 59995949 (GAD bigger-dt, 18 tasks). Partial: dt=0.005 at 10/30/50 pm; rest pending.
- Job 60002542 (Sella traj-log, 3 configs × 1 noise, n=50). For Sella per-step displacement figure.
- Job 60003770 (Sella libdef IRC, 6 noise × n=100). To populate canonical Sella IRC row.